

Задача 1:

$$A = \begin{pmatrix} 2 & -4 \\ 3 & 5 \\ -1 & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 2 & 7 \\ -3 & -4 & 0 \\ 5 & 2 & 1 \end{pmatrix}$$

$$C = \begin{pmatrix} 6 & -3 & 9 \\ 4 & -5 & 2 \\ 8 & 1 & 5 \end{pmatrix}$$

$$A^T = \begin{pmatrix} 2 & 3 & -1 \\ -4 & 5 & 0 \end{pmatrix}$$

$$A^T C = \begin{pmatrix} 2 & 3 & -1 \\ -4 & 5 & 0 \end{pmatrix} \cdot \begin{pmatrix} 6 & -3 & 9 \\ 4 & -5 & 2 \\ 8 & 1 & 5 \end{pmatrix} = \begin{pmatrix} 2 \cdot 6 + 3 \cdot 4 + (-1) \cdot 8 & 2 \cdot (-3) + 3 \cdot (-5) + (-1) \cdot 1 & 2 \cdot 9 + 3 \cdot 2 + (-1) \cdot 5 \\ -4 \cdot 6 + 5 \cdot 4 + 0 \cdot 8 & -4 \cdot (-3) + 5 \cdot (-5) + 0 \cdot 1 & -4 \cdot 9 + 5 \cdot 2 + 0 \cdot 5 \end{pmatrix} = \begin{pmatrix} 16 & -22 & 19 \\ -4 & -13 & -26 \end{pmatrix}$$

$$B^T = \begin{pmatrix} 1 & -3 & 5 \\ 2 & -4 & 2 \\ 7 & 0 & 1 \end{pmatrix}$$

$$A^T B^T = \begin{pmatrix} 2 & 3 & -1 \\ -4 & 5 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & -3 & 5 \\ 2 & -4 & 2 \\ 7 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 \cdot 1 + 3 \cdot 2 + (-1) \cdot 7 & 2 \cdot (-3) + 3 \cdot (-4) + (-1) \cdot 0 & 2 \cdot 5 + 3 \cdot 2 + (-1) \cdot 1 \\ -4 \cdot 1 + 5 \cdot 2 + 0 \cdot 7 & -4 \cdot (-3) + 5 \cdot (-4) + 0 \cdot 0 & -4 \cdot 5 + 5 \cdot 2 + 0 \cdot 1 \end{pmatrix} = \begin{pmatrix} 1 & -18 & 15 \\ 6 & -8 & -10 \end{pmatrix}$$

$$2A^T B^T = \begin{pmatrix} 2 & -36 & 30 \\ 12 & -16 & -20 \end{pmatrix}$$

$$D = A^T C - 2A^T B^T = \begin{pmatrix} 16 & -22 & 19 \\ -4 & -13 & -26 \end{pmatrix} - \begin{pmatrix} 2 & -36 & 30 \\ 12 & -16 & -20 \end{pmatrix} =$$

$$= \begin{pmatrix} 16-2 & -22-(-36) & 19-30 \\ -4-12 & -13-(-16) & -26-(-20) \end{pmatrix} = \boxed{\begin{pmatrix} 14 & 14 & -11 \\ -16 & 3 & -6 \end{pmatrix}}$$

Задача 2:

$$3 \cdot \begin{pmatrix} x & 2 & 3 \\ -1 & y & 4 \end{pmatrix} + 2 \begin{pmatrix} 1 & 2 & -5 \\ 2 & -6 & 2 \end{pmatrix} = \begin{pmatrix} 8 & v & -1 \\ 1 & 6 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 3x & 6 & 9 \\ -3 & 3y & 12 \end{pmatrix} + \begin{pmatrix} 2 & 4 & -10 \\ 4 & -12 & 22 \end{pmatrix} = \begin{pmatrix} 8 & v & -1 \\ 1 & 6 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 3x+2 & 6+4 & 9-10 \\ -3+4 & 3y-12 & 12+22 \end{pmatrix} = \begin{pmatrix} 8 & v & -1 \\ 1 & 6 & 4 \end{pmatrix}$$

$$3x+2 = 8 \Rightarrow 3x = 6 \Rightarrow x = 2$$

$$10 = v \Rightarrow v = 10$$

$$3y-12 = 6 \Rightarrow 3y = 18 \Rightarrow y = 6$$

$$12+2z = 4 \Rightarrow 2z = -8 \Rightarrow z = -4$$

$$x = 2$$

$$y = 6$$

$$z = -4$$

$$v = 10$$

3agame 3:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 6 & p \\ 5 & 10 & q \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & x \cdot 1 & 3 \\ 3 & x \cdot 3 & p \\ 5 & x \cdot 5 & q \end{pmatrix} \Rightarrow x = 2$$

$$\begin{matrix} 1 & 2 & 3 \\ 3 & 6 & p \\ 5 & 10 & q \end{matrix} \Rightarrow \begin{matrix} 3/1 = 3 \\ 6/2 = 3 \\ p/3 = 3 \end{matrix} \Rightarrow p = 9$$

$$5/1 = 5$$

$$\Rightarrow 10/2 = 5$$

$$q/3 = 5 \Rightarrow q = 15$$

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 6 & 9 \\ 5 & 10 & 15 \end{pmatrix}$$

Задание 4:

$$a_1 = (2, -5)^T \quad a_2 = (-1, 3)^T \quad x = (1, -4)^T$$

а) $x = \begin{pmatrix} 1 \\ -4 \end{pmatrix} \quad [x]_B = ?$

$$x = [x]_1 a_1 + [x]_2 a_2$$

$$\begin{pmatrix} 1 \\ -4 \end{pmatrix} = c_1 \begin{pmatrix} 2 \\ -5 \end{pmatrix} + c_2 \begin{pmatrix} -1 \\ 3 \end{pmatrix}$$

$$\begin{cases} 2c_1 - c_2 = 1 \\ -5c_1 + 3c_2 = -4 \end{cases}$$

$$c_2 = 2c_1 - 1$$

$$-5c_1 + 3(2c_1 - 1) = -4$$

$$-5c_1 + 6c_1 - 3 = -4$$

$$c_1 = 3 = -b$$

$$c_1 = -1$$

$$c_2 = -2 - 1$$

$$c_2 = -3$$

$$[x]_B = \begin{pmatrix} -1 \\ -3 \end{pmatrix}$$

б) $[y]_{\alpha} = ? \quad [y]_B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$$y = 1a_1 + 1a_2$$

$$y = \begin{pmatrix} 2 \\ -5 \end{pmatrix} + \begin{pmatrix} -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$\underline{[y]_{\alpha} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}}$$

Mat. analysis

3. Aufgabe 1:

$$f(x) = x_1^3 - 2x_1x_2 + x_2^2 - 3x_1 - 2x_2, x \in \mathbb{R}^2$$

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2} \right)$$

$$\frac{\partial f}{\partial x_1} = 3x_1^2 - 2x_2 - 3$$

$$\frac{\partial f}{\partial x_2} = -2x_1 + 2x_2 - 2$$

$$\nabla f(x) = (3x_1^2 - 2x_2 - 3, -2x_1 + 2x_2 - 2)$$

$$\begin{cases} 3x_1^2 - 2x_2 - 3 = 0 \\ -2x_1 + 2x_2 - 2 = 0 \end{cases} \Rightarrow \begin{cases} -2x_1 + 2x_2 - 2 = 0 \\ 2x_2 = 2x_1 + 2 \\ x_2 = x_1 + 1 \end{cases}$$

$$3x_1^2 - 2(x_1 + 1) - 3 = 0$$

$$3x_1^2 - 2x_1 - 5 = 0$$

$$x_1 = \frac{2 \pm \sqrt{4 + 60}}{6} = \frac{2 \pm 8}{6} = \frac{10}{6} = \frac{5}{3} \quad (\Rightarrow) \quad \frac{-6}{6} = -1$$

$$\text{bzw. } x_1 = \frac{5}{3} \quad x_2 = \frac{8}{3}$$

$$x_1 = -1 \quad x_2 = 0$$

Задача 2:

$$f(x_1, x_2) = \ln(\sqrt{x_1} + \sqrt{x_2}) \quad \text{yгбн.} \quad x_1 \frac{\partial f}{\partial x_1} + x_2 \frac{\partial f}{\partial x_2} = \frac{1}{2}$$

$$u = \sqrt{x_1} + \sqrt{x_2}, \quad f = \ln u$$

$$\frac{\partial f}{\partial x_1} = \frac{1}{u} \cdot \frac{\partial u}{\partial x_1} \quad \frac{\partial u}{\partial x_1} = \frac{1}{2\sqrt{x_1}}$$

$$\frac{\partial f}{\partial x_1} = \frac{1}{\sqrt{x_1} + \sqrt{x_2}} \cdot \frac{1}{2\sqrt{x_1}}$$

$$\frac{\partial f}{\partial x_2} = \frac{1}{\sqrt{x_1} + \sqrt{x_2}} \cdot \frac{1}{2\sqrt{x_2}}$$

$$x_1 \frac{\partial f}{\partial x_1} + x_2 \frac{\partial f}{\partial x_2} = x_1 \cdot \frac{1}{\sqrt{x_1} + \sqrt{x_2}} \cdot \frac{1}{2\sqrt{x_1}} + x_2 \cdot \frac{1}{\sqrt{x_1} + \sqrt{x_2}} \cdot \frac{1}{2\sqrt{x_2}}$$

$$x_1 \cdot \frac{1}{2\sqrt{x_1}} = \frac{x_1}{2\sqrt{x_1}} = \frac{\sqrt{x_1}}{2} \quad x_2 \cdot \frac{1}{2\sqrt{x_2}} = \frac{\sqrt{x_2}}{2}$$

$$x_1 \frac{\partial f}{\partial x_1} + x_2 \frac{\partial f}{\partial x_2} = \frac{1}{\sqrt{x_1} + \sqrt{x_2}} \left(\frac{\sqrt{x_1}}{2} + \frac{\sqrt{x_2}}{2} \right)$$

$$\frac{\sqrt{x_1}}{2} + \frac{\sqrt{x_2}}{2} = \frac{1}{2} (\sqrt{x_1} + \sqrt{x_2})$$

$$\frac{1}{\sqrt{x_1} + \sqrt{x_2}} \cdot \frac{1}{2} (\sqrt{x_1} + \sqrt{x_2}) = \frac{1}{2}$$

Вывод: функция удовлетворяет данному уравнению

Задача 3:

$$f(x, y, z) = (x + y + z, xyz)^T, \text{ точка } U = (1, 2, 3)^T$$

$$f_1(x, y, z) = x + y + z \quad f_2(x, y, z) = xyz$$

$$J_f = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \end{pmatrix}$$

$$\frac{\partial f_1}{\partial x} = 1 \quad \frac{\partial f_1}{\partial y} = 1 \quad \frac{\partial f_1}{\partial z} = 1$$

$$\frac{\partial f_2}{\partial x} = yz \quad \frac{\partial f_2}{\partial y} = xz \quad \frac{\partial f_2}{\partial z} = xy$$

$$J_f(x, y, z) = \begin{pmatrix} 1 & 1 & 1 \\ yz & xz & xy \end{pmatrix}$$

$$b) x = 1, y = 2, z = 3$$

$$yz = 2 \cdot 3, \quad xz = 1 \cdot 3 = 3, \quad xy = 1 \cdot 2 = 2$$

$$J_f(1, 2, 3) = \begin{pmatrix} 1 & 1 & 1 \\ 6 & 3 & 2 \end{pmatrix}$$

3 Aufgabe 4:

$$f(x) = \frac{1}{3} \|x\|_2^3 \quad x \in \mathbb{R}^n \setminus \{0\}$$

$$\|x\|_2 = \sqrt{x_1^2 + \dots + x_n^2} = (x^T x)^{1/2}$$

$$r = \|x\|_2$$

$$f(x) = \frac{1}{3} r^3$$

$$df(x) = r^2 dr$$

$$dr = \frac{1}{2} (x^T x)^{-1/2} \cdot 2 x^T dx = \frac{x^T dx}{r}$$

$$df(x) = r^2 \cdot \frac{x^T dx}{r} = r \cdot x^T dx = \|x\|_2 \cdot x^T dx$$

$$df(x) = (\nabla f(x))^T dx,$$

$$(\nabla f(x))_i = \frac{\partial f}{\partial x_i} = \|x\|_2 x_i$$

$$\nabla f(x) = \|x\|_2 x$$

Задача 5:

$$f(x) = \|x\|_2$$

$$\frac{\partial f}{\partial x_i} = \frac{1}{2} \left(\sum_{k=1}^n x_k^2 \right)^{-1/2} \cdot 2x_i = \frac{x_i}{\left(\sum_{k=1}^n x_k^2 \right)^{1/2}}$$

$$\frac{\partial f}{\partial x_i} = \frac{x_i}{\|x\|_2}$$

$$\nabla f(x) = \frac{x}{\|x\|_2}$$

$$df(x) = \sum_{i=1}^n \frac{x_i}{\|x\|_2} dx_i$$

$$df(x) = \frac{x \cdot dx}{\|x\|_2}$$

3agame 6:

$$f(x) = \|Ax\|_2, \quad x \in \mathbb{R}^n \setminus \{0\}, \quad A \in \mathbb{R}^{m \times n}$$

$$y = Ax \in \mathbb{R}^m$$

$$f(x) = \|y\|_2 = \sqrt{y_1^2 + y_2^2 + \dots + y_m^2} = \sqrt{(Ax)^T (Ax)}$$

$$h(x) = (Ax)^T (Ax) = x^T A^T A x$$

$$f(x) = h(x)^{1/2}$$

$$df(x) = \frac{1}{2} h(x)^{-1/2} \cdot dh(x), \quad h(x) > 0$$

$$h(x) = x^T M x, \quad M = A^T A$$

$$dh(x) = 2x^T M dx$$

$$x^T M dx + (Mx)^T dx = (x^T M + x^T M^T) dx$$

$$dh(x) = 2x^T M dx = 2x^T A^T A dx$$

$$df(x) = \frac{1}{2 \sqrt{x^T A^T A x}} \cdot [2x^T A^T A dx]$$

$$df(x) = \frac{x^T A^T A}{\sqrt{x^T A^T A x}} dx$$

$$df(x) = \frac{x^T A^T A}{\|Ax\|_2} dx$$

$$\nabla f(x)^T = \frac{x^T A^T A}{\|Ax\|_2}$$

$$\nabla f(x) = \frac{A^T A x}{\|Ax\|_2}$$

3. Aufgabe 7:

$$f(x) = -e^{-x^T x}, \quad x \in \mathbb{R}^n$$

$$df = \frac{\partial f}{\partial u} \cdot du$$

$$\frac{\partial f}{\partial u} = e^{-u} = e^{-x^T x}$$

$$du = d(x^T x) = 2x^T dx$$

$$df(x) = e^{-x^T x} \cdot (2x^T dx)$$

$$df(x) = 2e^{-x^T x} \cdot (x^T dx)$$

$$\nabla f(x) = 2e^{-x^T x} x$$

~~$\nabla f(x) = 2e^{-x^T x}$~~